

ConvNeXt-2U: A 3-D Deep Learning-Based Segmentation Model for Unified and Automatic Segmentation of Lungs, Normal Liver and Tumors in Y-90 Radioembolization Dosimetry

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Abstract—Y-90 radioembolization (RE) is an effective treatment for inoperable liver tumors. Pretreatment planning using Tc-99m-macroaggregated albumin (MAA) SPECT/CT requires segmentations of lung, normal liver and tumor, which could be delineated on low dose CT (LDCT), CT arterial portography (CTAP) and CT hepatic arteriography (CTHA). This study aims to develop a deep learning-based method for automatic lung, normal liver, and tumor segmentation for Y-90 RE treatment planning. Sixty-four sets of Tc-99m-MAA SPECT/CT, CTAP and CTHA images were retrospectively collected. Ground truth maps were provided by an experienced radiologist. We proposed ConvNeXt-2U, utilizing two U-Nets with connected skip connections and 3-D ConvNeXt blocks for joint segmentations. The LDCT, CTAP and CTHA were input to the two U-Nets. U-Net, attention U-Net, ResU-Net, MedNeXt, UNETR and Swin-UNETR were implemented for comparison. The segmentation performance was evaluated using Dice, Hausdorff distance (HD)95% and volume similarity (VS), and Y-90 RE dosimetrics, i.e., tumor-to-normal-liver ratio, lung-shunt fraction (LSF), absorbed dose (AD) of lungs, normal liver and tumors, and injected activity (IA). ConvNeXt-2U achieved the best performance in all segmentation indices and dosimetrics, except for HD95% of normal liver. It achieved mean Dice of 0.99, 0.93 and 0.77 in lungs, normal liver and tumors. ConvNeXt-2U provides a one-stop platform for unified segmentations for Y-90 RE treatment planning.

Index Terms—ConvNeXt, CT arterial portography (CTAP), CT hepatic arteriography (CTHA), image segmentation, Y-90 radioembolization (RE).

I. INTRODUCTION

Y-90 MICROSPHERES radioembolization (RE) is an effective treatment for inoperable liver cancer and

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metastatic tumors, which selectively damages tumor cells within close proximity by β -radiation emitted from Y-90 [1]. Patient selection in Y-90 RE and the corresponding IA depend on parameters determined from pretreatment Tc-99m-MAA SPECT or planar imaging, such as LSF, tumor-to-normal liver ratio (TNR), where delineating lung, liver, and tumor contours [2] are needed. In current clinical practice, the segmentation of these volumes-of-interest (VOI) requires substantial clinical efforts from radiologists. Therefore, the development of an automated tool for segmenting the lungs, liver, and tumors is in demand.

In recent years, deep learning algorithms, particularly convolution neural networks, have shown remarkable capabilities in automatic segmentation for medical imaging [3]. Numerous developments have been brought since the rise of U-Net architectures, and emerging trends of this field focus on conditional generative adversarial networks, cascaded networks and transformers [4]. Meanwhile, multimodal medical data for deep learning can effectively extract and integrate information of different modalities, improving clinical applicability [5], [6].

Among the segmentations for different organs [5], the segmentation of the lungs is a relatively easy task with exceptionally high precision [7]. On the other hand, liver and tumor segmentation is more challenging. Open datasets such as the liver tumor segmentation benchmark (LiTS) [8], consisting of portal venous phase contrast-enhanced CT (CECT) images, provide a platform for evaluating the performance of liver and liver tumor segmentation. U-Net [9], attention-based models [10], transformer-based models [11], [12], and ConvNeXt-based models [13] have progressively demonstrated improved performance in liver tumor segmentation. Notably, ConvNeXt, utilizing depth-wise convolution, layer normalization and inverted bottleneck by adopting the structure in transformer-based model [12], has addressed the performance limitations of transformer-based models in small datasets and has further demonstrated superior performance in larger datasets [14].

Two phases CECT, i.e., CTAP and CTHA, are commonly employed before Tc-99m-MAA SPECT to assess the tumor characteristics and to determine the liver segments for treatment [15]. On the other hand, LDCT from SPECT/CT can be used for lung and liver segmentation. CTAP, CTHA and

LDCT images could be used to extract the lung, normal liver, and tumor jointly for the dosimetry assessment of Y-90 RE. While existing works segment lungs, normal liver and tumors using different networks, in this study, we propose a novel 1-stop 3-D ConvNeXt backbone-based segmentation network (ConvNeXt-2U) for the joint segmentation of the lungs, normal liver and tumors from LDCT, CTAP, and CTHA images. We innovatively used a two-branch U-Net to segment these 3 VOIs from 3 different input CT images collectively. We utilized concatenated skip connections to share the processed information by the encoder from different input images, and introduced the use of a 3-D ConvNeXt module to enhance the segmentation performance. The segmentation performance as well as the dosimetric accuracy of the new network were compared with other standard network models.

The organization of this article is as follows. Section II describes the patient data, proposed and baseline networks and evaluation indices. Section III illustrates the results of the proposed and other baseline networks. Sections IV and V discuss the performance of the proposed network and summarize the findings.

II. RELATED WORKS

There are existing deep learning-based segmentation methods for Y-90 RE. Li et al. [16] and Tang et al. [17] examined liver segmentation from single-phase CECT images for Y-90 RE. Luu et al. [18] assessed a neural network for the segmentation of the lungs and liver from LDCT and single-phase CECT images to evaluate the LSF. Chaichana et al. [19] employed three different networks for the segmentation of tumors on SPECT images and the segmentation of the lungs and liver on LDCT images, respectively. However, segmentation accuracy based on SPECT is limited due to its inferior spatial resolution, while the clinical information provided by a single-phase CECT is also limited to discriminate tumors and other benign conditions. Moreover, utilization of multiple networks leads to a complex training and operational workflow. On the other hand, multiphase MRI [20] or multiphase CECT [21] could provide more precise liver and tumor contours.

III. MATERIALS AND METHODS

A. Patient Data Acquisition and Preprocessing

Fifty-seven sets of patient data, including Tc-99m-MAA SPECT/CT, CTAP, and CTHA scans for each patient, were retrospectively collected at Taipei Veterans General Hospital under the local ethics approval. Seven patients received contrast enhancement in both the left hepatic artery and the right hepatic artery, resulting in two sets of CTHA images. Thus, a total of 64 cases comprising Tc-99m-MAA SPECT/CT, CTAP, and CTHA data were included in the analysis, with two representative examples shown in Fig. 1. The patient characteristics is listed in Table I.

The SPECT image voxel size was $4.42 \times 4.42 \times 4.42 \text{ mm}^3$ and the matrix size was $128 \times 128 \times 128$. Corresponding LDCT image voxel size was $0.98 \times 0.98 \times 3.75 \text{ mm}^3$ with a matrix size of $512 \times 512 \times 108$. All patients were administered with 111 MBq Tc-99m-MAA to simulate the

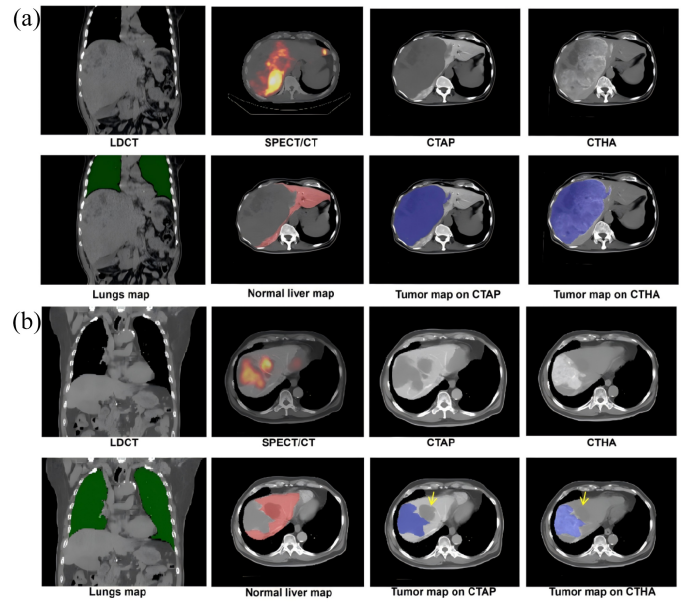


Fig. 1. Sample LDCT, SPECT/CT, CTAP and CTHA images (top) and corresponding lung (green), normal liver (red), and tumor (blue) maps (bottom) manually segmented by a radiologist for 2 patients. The normal liver is enhanced on CTAP, and the tumor is enhanced on CTHA. (a) Patient with a tumor well enhanced only on CTHA while normal liver well enhanced only on CTAP. (b) Patient with a cyst not enhanced on both CTAP and CTHA (yellow arrows).

TABLE I
PATIENT DEMOGRAPHICS IN THIS STUDY

Characteristics	Mean $\pm$ STD [min, max]
Age (years)	65.9 $\pm$ 13.9 [25.0, 89.0]
Gender	41:16 (male: female)
Height (m)	1.61 $\pm$ 0.39 [1.43, 1.73]
Weight (kg)	64.8 $\pm$ 12.1 [48.0, 90.4]
BMI (kg/m ²)	25.2 $\pm$ 3.0 [21.6, 32.0]
Right : Left hepatic artery	54:10:00
Tumor type	
Hepatocellular carcinoma	54
Neuroendocrine tumor	2
Metastatic colorectal cancer	1
Tumor volume (mL, n=355)	95.2 $\pm$ 366.5 [1.8, 3662.8]
Diameter > 2 cm	n=66
Diameter > 5 cm	n=29

Note: Numerical data were presented in mean $\pm$ SD and [range].

Y-90 microspheres distribution, and the detailed SPECT/CT imaging protocol was described in a published literature [2]. For the acquisition of CECT scans, arterial vascular access was established by placing a catheter in the superior mesenteric artery for CTAP, while another in the common hepatic artery for CTHA, respectively. The scan time was set to be 35–42 s for CTAP and 12–20 s for CTHA post contrast injection, depending on patients' actual situation.

In CTAP, Ultravist 370 (Bayer Vital Leverkusen, Germany) was injected at a rate of 2.0–2.5 mL/s, and CT scans were performed 40 s after injection. In CTHA, the same contrast agent was injected at a rate of 0.5–1.2 mL/s, and CT scans were conducted 8 s after injection. The CTAP and CTHA scans, performed at 120 kVp and 515 mA, were reconstructed with a matrix size of $512 \times 512 \times (29-53)$ and a voxel size of $(0.54-0.79) \times (0.54-0.79) \times 5.00 \text{ mm}^3$.

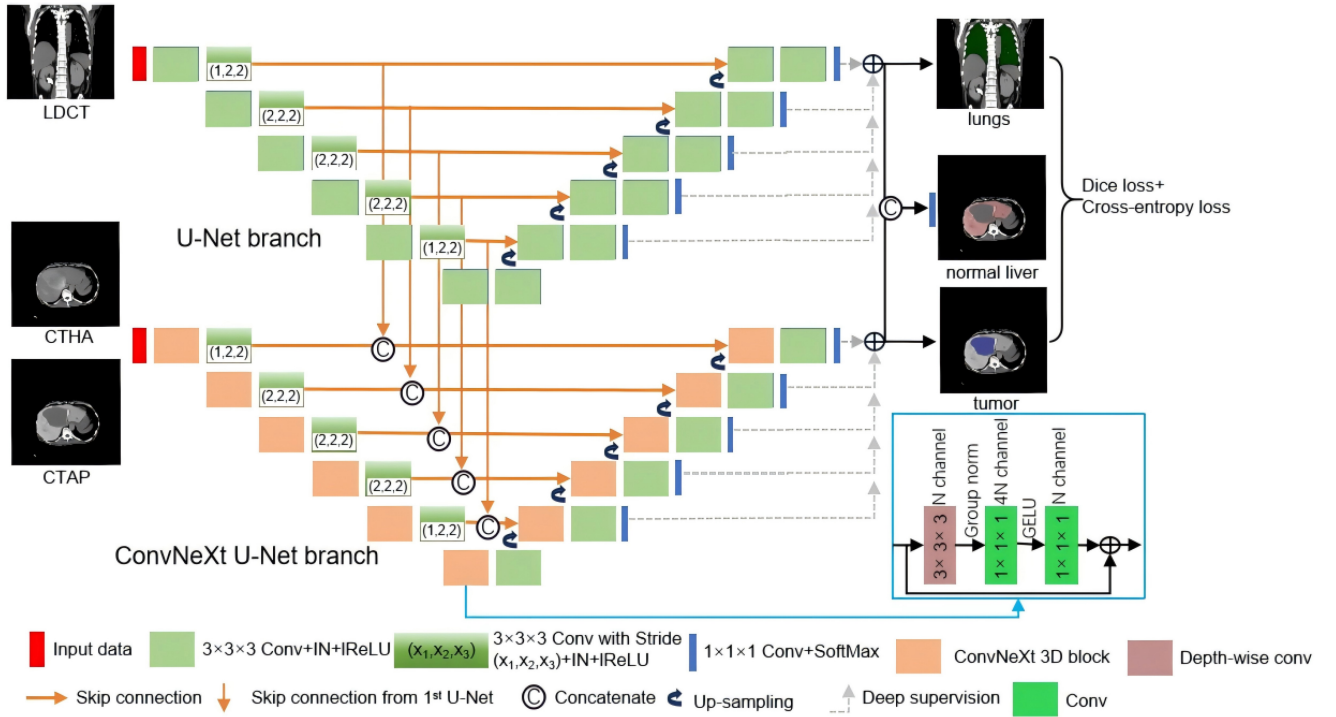


Fig. 2. Proposed ConvNeXt-2U using two U-nets with a 3-D ConvNeXt block on 1 branch.

The CTHA images were aligned to the CTAP images by rigid+b-spline transformation [22]. LDCT and CTAP were initially manually aligned, and then rigid +b-spline deformation was performed using coarse liver masks of two modalities to register CTAP to LDCT [23]. The transformation matrix was subsequently applied to CTHA to ensure the alignment of three CT images. All registrations were performed using Elastix [24]. The reference tumor maps were segmented on co-registered CTAP and CTHA images by a radiologist with 15-year experience. The normal liver map was acquired by excluding tumors in the whole liver map which was obtained from LDCT. A total of 355 tumors were segmented from the 57 patients. The lung maps were delineated using a threshold of -400 HU [25] and manually fine-tuned on LDCT to exclude trachea and bronchioles, which also exhibit low HU values within the lungs, as the reference. All contours were then assessed by the whole Y-90 RE team for final approval. Other preprocessing, including resampling, normalization and data split, was performed under the nnU-Net pipeline [26].

B. Proposed Network Architecture

The proposed network, ConvNeXt-2U, consisted of two branches with U-Net architecture is shown in Fig. 2. The first U-Net branch adopted the nnU-Net network automatic configuration and took LDCT as input and produced segmentations for the lungs and normal liver. The second U-Net branch, enhanced with a 3-D ConvNeXt module to replace the first convolution module in each layer, employed co-registered CTAP and CTHA as input and produced segmentations for the normal liver and tumors. The 3-D ConvNeXt module consisted of a depth-wise convolutional layer utilizing a $3 \times 3 \times 3$ kernel and a stride of 1, followed by a group normalization,

then a $1 \times 1 \times 1$ convolution layer to increase the number of channels to four times of the input channels, incorporating a Gaussian error linear unit (GELU) activation function. Finally, another $1 \times 1 \times 1$ convolution layer was used to map the number of channels back to the original number of input channels, with a residual connection linking the input and output. Both branches incorporated a $1 \times 1 \times 1$ convolution layer and SoftMax activation to implement deep supervision in the decoder [27]. The skip connections of the first U-Net branch were concatenated to the corresponding skip connections in the second U-Net branch to share the lung and liver information. Thus, the normal liver segmentation result was jointly provided by 2 branches. An ablation study was conducted to evaluate the effectiveness of each component. The network's output was compared to the reference maps to calculate the sum of Dice and cross-entropy as the training loss function

$$\text{Loss} = -\frac{2}{A} \sum_{a \in A} \frac{\sum_{i \in I} P_i^a T_i^a}{\sum_{i \in I} P_i^a + \sum_{i \in I} T_i^a} + \frac{1}{N} \sum_{i \in I} \sum_{a \in A} T_i^a \log P_i^a \quad (1)$$

where the first part is Dice and the second part is cross entropy. P is the predicted value for each class, T is the one-hot encoding of the ground truth segmentation, and I is the image size. A and N are the total number of classes and voxels, respectively, where a and i iterate over all classes and voxels.

C. Segmentation Network Comparison and Training

The proposed method was compared to other baseline segmentation networks, including 3-D U-Net [28], Attention U-Net [29], ResU-Net [30], ConvNeXt-based

MedNeXt-B [13], transformer-based UNETR [11] and Swin-UNETR [12]. U-Net, Attention U-Net and ResU-Net are consisted of six layers, while MedNeXt-B, UNETR and Swin-UNETR utilized the default architecture without modification. MedNeXt-B used a kernel size of 3 due to the hardware constraint and training time limitation. All baseline networks were implemented with a single branch under nnU-Net framework. The inputs of these networks were LDCT, CTHA and CTAP all together, and the output were lung, normal liver and tumor maps. The performance of all networks including those mentioned in the ablation study previously, was evaluated on our datasets through a fourfold cross-validation, with a training, validation, and testing ratio of 44:4:16. All networks, were trained using nnU-Net's default settings as in ConvNeXt-2U, including a batch size of 2, 1000 epochs and data augmentation. Stochastic Gradient Descent optimization was utilized with an initial learning rate of 0.01 and Nesterov momentum ($\mu = 0.99$) [26]. All training processes were performed on a local server with a Nvidia A6000 GPU.

D. Data Analysis and Dosimetric Evaluation

The segmentation performance of different networks was first compared using dice coefficient (Dice), HD95% and VS for target VOIs (lungs, normal liver, and tumors). The Dice measures volume overlap between the segmentation result and the ground truth

$$\text{Dice} = \frac{2 \times |V_{\text{network output}} \cap V_{\text{ground truth}}|}{|V_{\text{network output}}| + |V_{\text{ground truth}}|} \quad (2)$$

where $V_{\text{network output}}$ is the segmentation volume generated by different network output and $V_{\text{ground truth}}$ is the segmentation volume of the ground truth.

HD95% measures the 95% of maximum distance between the surface of the segmentation result and the surface of the ground truth

$$\text{HD95\%} = \max_{95\%}[d(X, Y), d(Y, X)] \quad (3)$$

where X is the different network output and Y is the ground truth, $d(X, Y)$ is the minimum distance between the surface of X and Y , $\max_{95\%}$ is the 95th percentile of the maximum value.

VS evaluates the VS between the segmentation result and ground truth

$$\text{VS} = 2 \times \frac{|V_{\text{network output}}| - |V_{\text{ground truth}}|}{|V_{\text{network output}}| + |V_{\text{ground truth}}|} \quad (4)$$

Y-90 RE treatment planning indices, including LSF, TNR, AD of target VOIs, are all obtained from Tc-99-MAA SPECT images based on the CT segmentations. These indices are compared based on the segmented volumes of different networks and the reference segmentation to assess their clinical relevance.

LSF is used to estimate the microspheres shunting to the lungs caused by Y-90 RE

$$\text{LSF} = 100\% \times \frac{A_{\text{lungs}}}{A_{\text{lungs}} + A_{\text{liver}}} \quad (5)$$

where A_{lungs} and A_{liver} is the activity within lungs and whole liver (normal liver+tumor maps), respectively.

TNR measures the relative concentration or activity of a tumor compared to the surrounding normal liver tissue

$$\text{TNR} = \frac{A_{\text{tumors}}/M_{\text{tumors}}}{A_{\text{normal liver}}/M_{\text{normal liver}}} \quad (6)$$

where M_{tumors} and $M_{\text{normal liver}}$ are the tumor mass and the normal liver, respectively.

The AD of each VOI is calculated using the MIRD model [31]

$$\text{AD}_{\text{VOI}} = \frac{49.67 \times A_{\text{VOI}}}{M_{\text{VOI}}} \quad (7)$$

where 49.67 (J/GBq) is the constant for converting activities to AD for Y-90. IA is calculated as follows:

$$\text{IA} = \min \left[\frac{70(M_{\text{tumors}} \times \text{TNR} + M_{\text{normal}})}{49.67 \left(1 - \frac{\text{LSF}}{100}\right)}, \frac{30M_{\text{lungs}}}{49.67 \frac{\text{LSF}}{100}} \right] \quad (8)$$

where 30 Gy and 70 Gy is the upper AD limit of lungs and normal liver, respectively. All tumor related indices, e.g., physical indices, TNR, $\text{AD}_{\text{tumors}}$, $\text{AD}_{\text{normal liver}}$ and IA were assessed for tumors with all sizes as well as diameter >2 cm according to the EANM recommendation [1].

The Wilcoxon signed-rank test with Bonferroni correction was used for the statistical analysis between ConvNeXt-2U and baseline networks. A p-value <0.05 is considered statistically significant.

IV. RESULTS

A. Ablation Study

To validate the effectiveness of each component in our proposed network, we conducted an ablation study. We started with (i) a single U-Net and progressively extended it to (ii) two U-Nets, (iii) using shared skip connections, (iv) liver concatenation, and (v) using 3-D ConvNeXt blocks on the second branch. The 3-D ResNet blocks were also implemented to substitute the 3-D ConvNeXt on the second U-Net branch for comparison. We did not replace convolution blocks on the first branch with the ConvNeXt modules due to the limitation in memory for computing.

The segmentation results of the ablation study are presented in Table II. The two U-Net structures with shared skip connection or liver concatenation could both slightly improve the normal liver mean Dice from 0.92 to 0.93 and improve tumor mean Dice from 0.73 to 0.74 compared with single U-Net, respectively. The combination of shared skip connection and liver concatenation could further improve the segmentation performance in normal liver (mean Dice = 0.93) and tumors (mean Dice = 0.74). The utilization of 3-D ConvNeXt blocks had the best segmentation performance in all indices of normal liver and tumors followed by the use of ResNet blocks. All segmentation results were similar in lung segmentation. The comparison results demonstrated that the proposed components, i.e., shared skip connection, liver concatenation and 3-D ConvNeXt blocks, could contribute to the improvement of model performance for this segmentation task.

TABLE II
SEGMENTATION RESULTS OF THE ABLATION STUDY. THE BEST RESULTS ARE MARKED IN RED AND SECOND BEST ARE MARKED IN BLUE

Network combination						Normal liver			Lungs			Tumors per patient		
Single U-Net	Two U-Net	Shared skip conn	Liver concat	ConvNeXt block	ResNet block	Dice▲	VS▼	HD95%▼	Dice▲	VS▼	HD95%▼	Dice▲	VS▼	HD95%▼
✓						0.92±	0.07±	4.93±	0.99±	0.00±	0.98±	0.73±	0.41±	32.55±
	✓	✓				0.07	0.12	12.07	0.00	0.00	0.12	0.26	0.51	95.37
						0.93±	0.08±	4.48±	0.99±	0.00±	0.89±	0.74±	0.39±	31.23±
	✓		✓			0.07	0.12	10.15	0.00	0.00	0.31	0.26	0.50	95.30
						0.93±	0.08±	3.69±	0.99±	0.00±	0.93±	0.74±	0.38±	29.79±
	✓	✓	✓			0.07	0.11	6.76	0.00	0.00	0.24	0.26	0.50	94.58
						0.93±	0.08±	3.81±	0.99±	0.00±	0.89±	0.74±	0.39±	30.87±
	✓	✓	✓	✓		0.07	0.12	11.58	0.00	0.00	0.31	0.26	0.52	95.34
						0.93±	0.07±	3.43±	0.99±	0.00±	0.97±	0.77±	0.32±	19.73±
	✓	✓	✓		✓	0.07	0.11	6.65	0.00	0.00	0.17	0.22	0.43	35.70
						0.93±	0.08±	3.47±	0.99±	0.00±	0.90±	0.76±	0.36±	29.13±
	✓	✓	✓			0.07	0.12	6.39	0.00	0.00	0.29	0.24	0.46	94.62

Upward arrow signifies that a larger value means better performance; downward arrow signifies that a smaller value means better performance.

TABLE III
SEGMENTATION PERFORMANCE OF DIFFERENT NETWORKS. THE BEST RESULTS ARE MARKED IN RED AND SECOND BEST ARE MARKED IN BLUE

Mean±STD	Normal liver			Lungs			All tumors per patient			Tumors with diameter >2 cm per patient		
	Dice▲	VS▼	HD95%▼	Dice▲	VS▼	HD95%▼	Dice▲	VS▼	HD95%▼	Dice▲	VS▼	HD95%▼
U-Net	0.92±	0.09±	4.93±	0.99±	0.00±	0.98±	0.73±	0.41±	32.55±	0.79±	0.35±	35.57±
	0.07*	0.12	12.07*	0.00*	0.00*	0.12	0.26*	0.51*	95.37*	0.23*	0.49*	135.75*
attention U-Net	0.92±	0.09±	5.14±	0.99±	0.00±	1.02±	0.74±	0.39±	20.57±	0.78±	0.35±	22.88±
	0.08*	0.12	11.40*	0.00*	0.00*	0.13*	0.24*	0.46*	31.68*	0.22*	0.43*	97.66
ResU-Net	0.92±	0.09±	5.15±	0.99±	0.01±	1.74±	0.73±	0.41±	21.16±	0.78±	0.34±	24.09±
	0.08*	0.12*	11.39*	0.01*	0.01	5.84*	0.26*	0.50*	35.16*	0.25*	0.43*	98.63*
MedNeXt	0.93±	0.08±	3.43±	0.99±	0.00±	0.99±	0.75±	0.37±	28.63±	0.79±	0.33±	36.29±
	0.07	0.11*	6.46	0.00*	0.00*	0.17*	0.25	0.47*	93.84*	0.23*	0.45*	136.27*
UNETR	0.89±	0.12±	6.98±	0.99±	0.01±	2.24±	0.71±	0.44±	35.25±	0.73±	0.36±	26.87±
	0.13*	0.24	17.42*	0.01*	0.01*	11.84*	0.27*	0.57*	94.16*	0.24*	0.47*	98.2
Swin-UNETR	0.90±	0.11±	6.91±	0.99±	0.00±	1.00±	0.72±	0.42±	33.40±	0.78±	0.34±	36.94±
	0.10*	0.14	13.29*	0.00*	0.00*	0.05*	0.25*	0.49*	95.44*	0.23*	0.47*	136.15*
ConvNeXt-2U	0.93±	0.07±	3.43±	0.99±	0.00±	0.97±	0.77±	0.32±	19.73±	0.83±	0.24±	21.47±
	0.07	0.11	6.65	0.00	0.00	0.17	0.22	0.43	35.70	0.17	0.35	97.78

*p<0.05; upward arrow signifies that a larger value means better performance; downward arrow signifies that a smaller value means better performance.

B. Segmentation Performance for Physical Indices

The segmentation performance of different networks is shown in Table III, and a sample patient result is shown in Fig. 3. For the normal liver segmentation, except for the transformer-based UNETR and SwinUNETR, which performed relatively poorly, the other methods showed comparable performance with Dice >0.9 and VS <0.1. MedNeXt had the best performance in HD95% followed by ConvNeXt-2U. U-Net and attention U-Net ranked lower than the proposed ConvNeXt-2U network and MedNeXt, while better than Swin-UNETR. The results from networks closely aligned with the ground truth with mean Dice >0.9 in Fig. 3(a). Only MedNeXt had no statistical difference with ConvNeXt-2U in normal liver Dice and HD95%.

For lung segmentation, all networks had achieved segmentation performance with a mean Dice >0.99 [Fig. 3(b)]. The proposed network achieved slight improvement than others in all indices. All networks had statistical difference with ConvNeXt-2U in lung Dice.

For tumor segmentation, all networks had worse performance than lung and normal liver segmentation, with a mean Dice <0.8. Among them, ConvNeXt-2U outperformed others, achieving mean Dice of 0.77, mean VS of 0.32, and mean HD95% of 19.73. MedNeXt with mean Dice of

0.75 and mean VS of 0.37, and Attention U-Net with mean HD95% of 20.57 ranked second. Swin-UNETR had the worst performance in all indices. The tumor segmentation results of a sample patient are shown in Fig. 3(c). MedNeXt had no statistical difference with ConvNeXt-2U in tumor Dice, while all other methods had statistical difference with ConvNeXt-2U.

The segmentation performance of networks for tumors with diameter >2 cm per patient is also shown in Table III. ConvNeXt-2U also exhibited the best performance with mean Dice of 0.83, mean VS of 0.24 and mean HD95% of 21.47. MedNeXt ranked second with mean Dice of 0.79, mean VS of 0.33 and attention U-Net ranked second with mean HD95% of 22.88, consistent with the results of the all tumor cases. U-Net and attention U-Net also achieved better performance than Swin-UNETR. All networks demonstrated better performance in larger tumors and worse performance in smaller ones, as illustrated in Fig. 4. All baseline networks had statistical difference with ConvNeXt-2U in tumors with diameter >2 cm Dice and VS.

The execution time for training processes of U-Net, attention U-Net, MedNeXt, Swin-UNETR and ConvNeXt-2U was 43, 46, 45, 55, 58, 61 and 53 h, respectively, while their prediction time per patient was 61, 65, 63, 73, 80, 85 and 71 s, respectively.

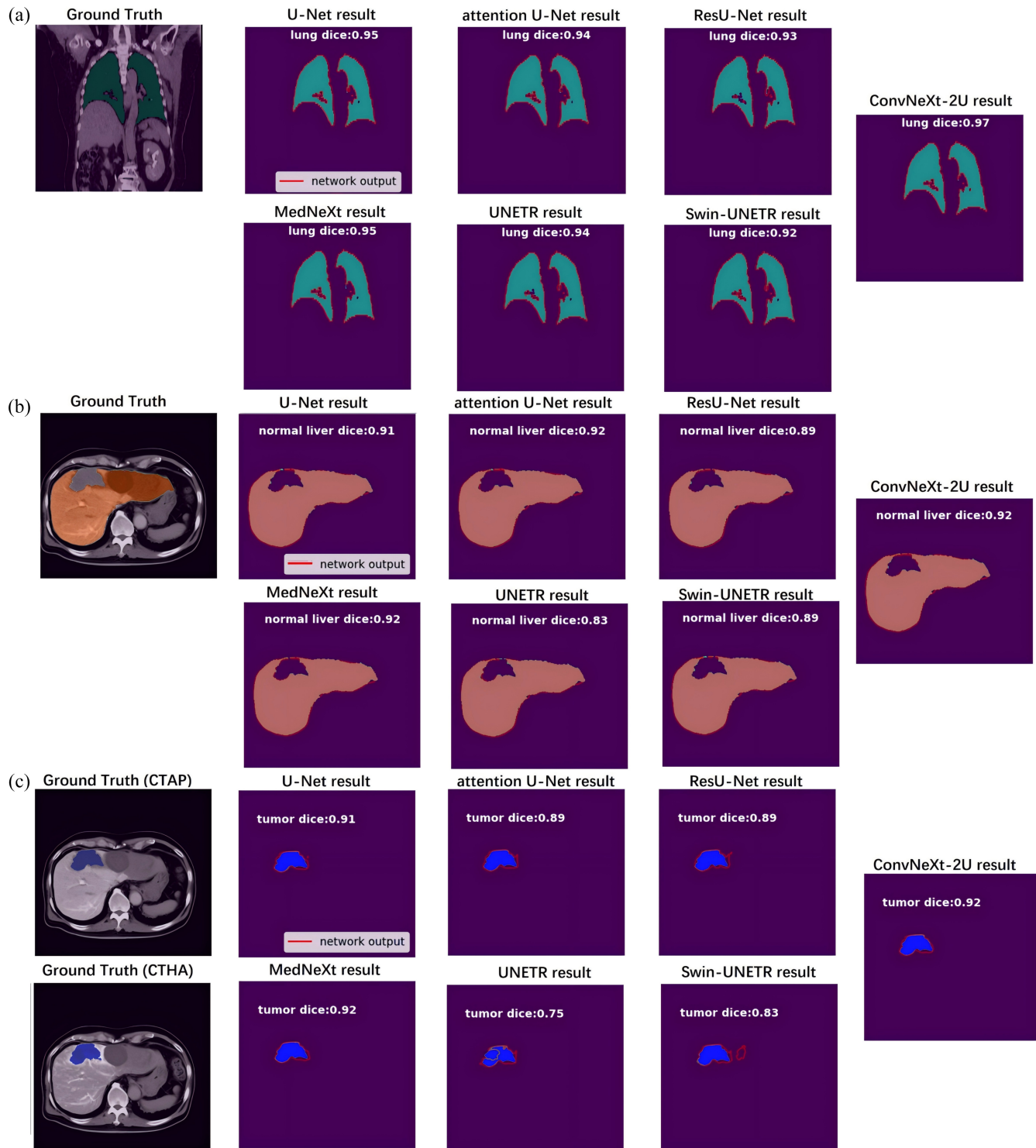


Fig. 3. Sample patient segmentation results for (a) lungs (green), (b) normal liver (red) and (c) tumors (blue) for each network.

C. Dosimetric Evaluation

The dosimetric errors in Y-90 RE treatment planning from different segmentation errors are presented in Table IV. All IAs are calculated according to the upper AD limit of the normal liver. The proposed ConvNeXt-2U exhibited the best performance among all methods. U-Net ranked second in lung-related indices, i.e., LSF mean absolute error of 2.6%

and AD_{lungs} mean absolute error of 2.0%. MedNeXt ranked second in TNR mean absolute error of 27.7%, $AD_{\text{normal liver}}$ of 16.6% and IA error of 9.3%. Swin-UNETR had the worst performance except for AD_{lungs} . All baseline networks had statistical difference with ConvNeXt-2U in LSF and AD_{tumors} . UNETR and Swin-UNETR did not have statistical differences with ConvNeXt-2U in IA. The dosimetric errors

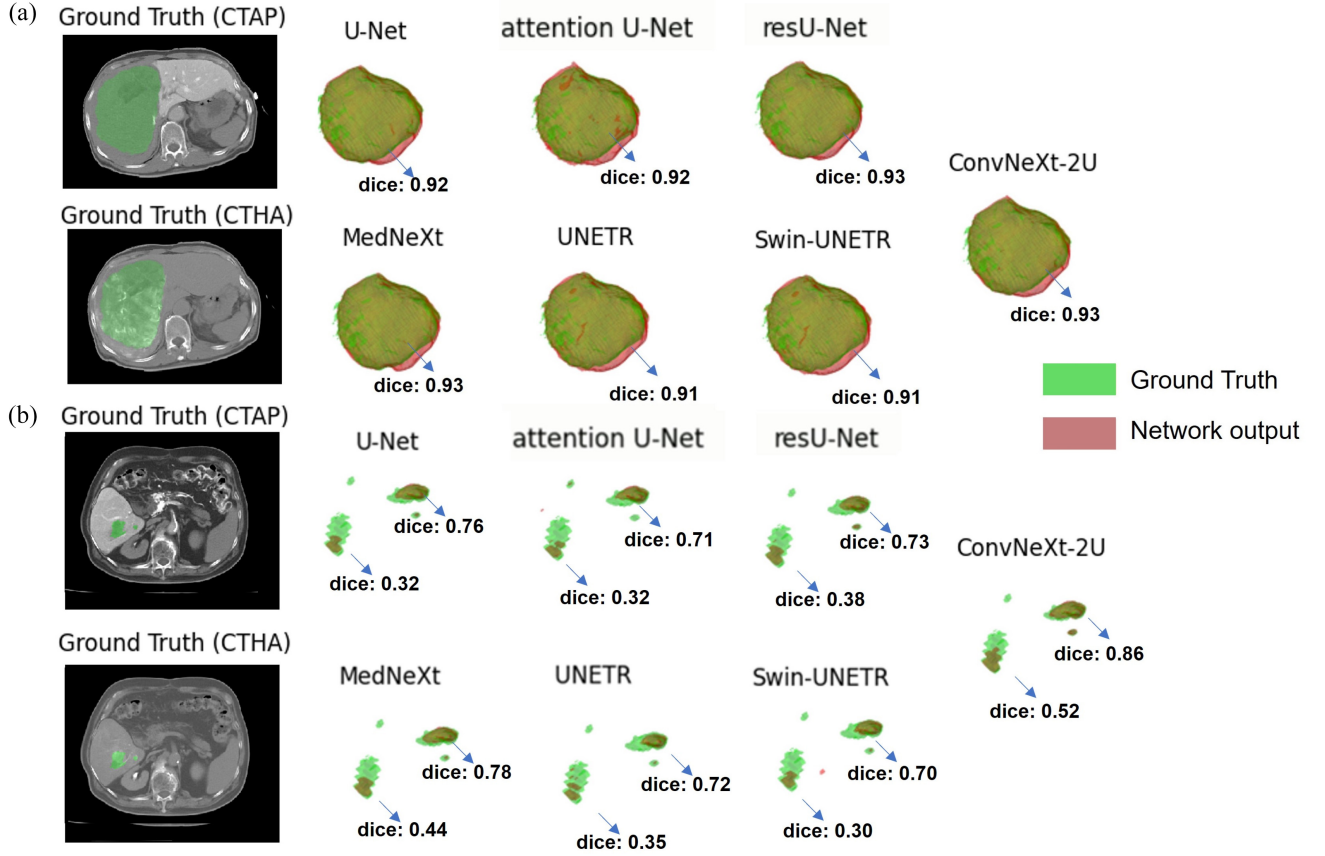


Fig. 4. Sample 3-D tumor segmentation results: (a) big tumor (diameter: 90 mm), (b) small tumors (diameter: 6-22 mm).

TABLE IV
ABSOLUTE ERRORS OF Y-90 RE TREATMENT PLANNING DOSIMETRICS BY DIFFERENT NETWORKS. THE BEST RESULTS ARE MARKED IN RED AND SECOND BEST ARE MARKED IN BLUE

Absolute error (%)	All tumors per patient						Tumors with diameter >2 cm per patient			
	LSF	TNR	$AD_{normal\ liver}$	AD_{lungs}	AD_{tumors}	IA	TNR	$AD_{normal\ liver}$	AD_{tumors}	IA
U-Net	2.6±	31.3±	13.1±	2.0±	17.9±	9.4±	19.2±	12.7±	9.4±	9.3±
	2.4*	69.1	26.5*	1.7*	37.5*	14.8*	28.7	25.6*	13.7	14.5*
attention U-Net	3.4±	29.8±	14.5±	2.9±	17.7±	10.5±	25.0±	14.8±	12.9±	10.8±
	3.5*	58.2*	29.1*	3.1*	34.7*	15.1*	50.0	29.9*	27.2*	15.2*
ResU-Net	3.3±	29.6±	14.1±	2.4±	17.7±	10.1±	23.8±	14.2±	12.5±	3.3±
	4.0*	64.7	27.8*	2.8	38.3*	15.0*	55.2	28.1*	34.3*	4.2*
MedNeXt	3.0±	27.7±	12.1±	2.4±	16.6±	9.3±	22.7±	12.3±	11.7±	9.6±
	3.1*	61.8*	24.0*	2.5*	34.4*	14.8*	48.2*	24.1*	28.5*	15.0
UNETR	6.6±	33.7±	18.2±	3.4±	22.0±	13.2±	24.9±	18.7±	12.7±	14.3±
	15.1*	54.9	34.3	5.0	40.3*	22.7	35.9*	35.8	16.4*	23.6*
Swin-UNETR	5.9±	32.9±	16.5±	2.1±	19.1±	12.0±	22.1±	17.1±	18.3±	12.7±
	16.4*	66.9	35.1	2.5*	41.1*	19.4	35.3	35.2	9.7*	19.9
ConvNeXt-2U	2.4±	23.5±	11.5±	2.0±	12.6±	8.4±	17.9±	11.3±	8.0±	8.3±
	2.2	49.9	27.6	1.7	21.5	14.3	34.3	27.1	14.3	14.1

* $p < 0.05$; upward arrow signifies that a larger value means better performance; downward arrow signifies that a smaller value means better performance.

of the tumors with diameter >2 cm (per patient) are also presented in Table IV. The errors of TNR and AD_{tumors} were smaller by excluding small tumors. The proposed ConvNeXt-2U also achieved the best performance with $AD_{normal\ liver}$ mean absolute error of 11.3%, AD_{tumors} mean absolute error of 8.0%, IA error of 8.3%, and TNR mean absolute error of 17.9%. The U-Net achieved second best performance in the mean absolute error of the TNR of 19.2%, AD_{tumors} error of 9.4% and IA mean absolute error of 9.3%. MedNeXt achieved second best performance in $AD_{normal\ liver}$ with mean absolute

error of 12.3%. MedNeXt and Swin-UNETR had no statistical difference with ConvNeXt-2U in IA.

V. DISCUSSION

In this study, we proposed a new ConvNeXt-2U network and conducted a comprehensive analysis of its segmentation performance in lungs, normal liver and tumors along with various other deep learning networks for Y-90 RE treatment planning. All networks were implemented based on the

nnU-Net framework [26], where both U-Net and Attention U-Net exhibited similar performance in normal liver and tumor segmentation tasks compared to results reported in the LiTS dataset [8]. However, the improvement in Attention U-Net over U-Net was somewhat limited under the nnU-Net framework [32]. Swin-UNETR, a Vision transformer-based method, had the worst performance, and similar results were observed in another lung and normal liver segmentation study [18]. Vision transformers achieved excellent performance on large datasets, such as the BraTS dataset consisting of 2000 patients for brain tumor segmentation [12]. However, the performance of transformers-based network tended to degrade for smaller datasets, while the ConvNeXt-based architecture, e.g., MedNeXt, outperformed convolution-based network and transformers on smaller datasets [13], [14], consistent with our study. Compared with single-branch networks, the two-branch network could integrate multimodal image information [33], [34]. Thus, our proposed ConvNeXt-2U exhibited the best performance in physical and clinical segmentation indices.

For the segmentation of the lungs and normal liver, all networks achieved results exceeding mean Dice of >0.99 and >0.9 , respectively, consistent with other literatures [8], [18], [35]. The lungs have lower density and distinct boundaries with other tissues, thus is easily segmented on CT images. However, it is necessary to remove low-HU regions within lungs, such as the trachea and bronchioles. Therefore, deep learning-based method is still more advantageous compared to the standard thresholding method, and the unified network acquired state-of-the-art results in lung segmentation. The normal liver segmentation is constrained by the challenging tumor segmentation as shown in Fig. 3. There was no mis-segmentation of surrounding organs or soft tissue observed in all patients, probably attributed to the enhanced contrast from CTAP and CTHA [21].

For tumor segmentation, the best mean Dice is only 0.77 by the proposed ConvNeXt-2U. The mean inter- and intra-observer mean Dice of liver tumors were 0.79 and 0.85, respectively, in a MRI-based segmentation study [36]. In the LiTS dataset, where only the low-density regions in the CECT were considered as tumors, state-of-the-art segmentation methods achieved a Dice score of 0.76 [8]. However, the tumor characteristics for patients enrolled for Y-90 RE treatment were usually more complex as compared to the LiTS dataset, where necrosis and cysts often existed. Therefore, only regions showing unenhanced intensity in CTAP and enhanced intensity in CTHA were considered as tumors in this study. The baseline nnU-Net achieved a mean Dice of 0.73, comparable with performance evaluated in the LiTS results. All networks show lower performance in segmenting small tumors compared to large tumors, which is consistent with findings in other literatures [37], [38]. One possible reason is that convolutional blocks perform multiple down-sampling operations on the images, which can easily lead to the loss of information on small structures. Therefore, specialized modules are needed to optimize the performance in segmenting small objects [39].

The effectiveness of each component, i.e., shared skip connection, liver concatenation, and 3-D ConvNeXt block, was demonstrated in the ablation study. The two-branch architecture enhances normal liver, tumor and lung segmentation by generating outputs for specific tasks in each branch. We assumed that the first branch, responsible for segmenting the lungs and entire liver, can leverage positional information about the normal liver and tumors obtained from the second branch. This strategy introduced new gradient back propagation paths, increasing information flow between two branches. Moreover, the ConvNeXt module potentially improves normal liver and tumor segmentation as compared to the ResNet and convention convolution blocks, as it utilizes depth-wise convolution, which is considered equivalent to self-attention and capable of capturing global information [40]. Consequently, it was applied in the first block of the decoder in the second branch to effectively process the information from the concatenated skip connection and capture global information.

For the dosimetric indices, our proposed ConvNeXt also showed the best performance over all dosimetrics. Regarding the LSF used for excluding potential radiation pneumonia in the clinics, the proposed network was consistent with physician's manual segmentation results [18]. All networks exhibited $>23\%$ errors in TNR and $>12\%$ errors in AD_{tumors} , due to the subpar tumor segmentation performance. For tumors with diameter > 2 cm, AD_{tumors} and TNR had fewer errors than those using all tumors as expected.

Accurate segmentation of lungs, normal liver, and tumors is pivotal for optimizing AD of critical organs and tumors, ensuring patient safety in Y-90 RE. The proposed ConvNeXt-2U can complete 3 segmentation tasks in one step for Y-90 RE to reduce the clinical burden. The two branches architecture can separate 3 different tasks, i.e., lung, normal liver and tumor segmentation, while the shared information between 2 branches enhances overall segmentation accuracy. The adaptation of the 3-D ConvNeXt block also successfully improved the network's accuracy. Thus, the proposed ConvNeXt-2U demonstrated better performance than six baseline networks, offering a streamlined solution for comprehensive Y-90 RE treatment planning.

However, there are certain limitations in this study. The use of the 3-D ConvNeXt block occupies a significant amount of computing memory. Therefore, it can only be implemented in 1 U-Net branch currently. The reference tumor maps were validated by a radiologist. Though the radiologist was with rich clinical experience (15 years) and contour were further endorsed by the whole Y-90 RE team, further study could include reference maps provided by multiple operators to enhance the generalizability of the segmentation model. Besides, the tumor type in this study is mainly hepatocellular carcinoma. More types of metastatic cancer need to be evaluated, as their imaging characteristics could be quite different from hepatocellular carcinoma. Last but not least, our evaluation was performed on a relatively small dataset. Further validation on larger and more diverse patient cohorts is warranted.

VI. CONCLUSION

We have developed and evaluated a unified ConvNeXt-2U network for automatic segmentation of lungs, normal liver and tumors for Y-90 RE treatment planning. The proposed network and baseline convolution-based networks achieved comparable and excellent performance in lungs and normal liver. The proposed network and baseline convolution-based networks achieved comparable and excellent performance in lungs (Dice >0.99, VS <0.1, and HD95% <3.0) and normal liver (Dice >0.89, VS <0.12, and HD95% <7.0). The ConvNeXt-2U network also achieved better segmentation performance in tumors compared with other baseline networks, with a mean Dice of 0.77, VS of 0.32 and HD95% of 19.73, respectively.

ABBREVIATION GLOSSARY

Abbreviation	Definition
RE	radioembolization
MAA	macroaggregated albumin
Dice	Dice coefficient
IA	injected activity
LSF	lung-shunt fraction
TNR	tumor-to-normal liver ratio
VOI	volumes-of-interest
CECT	contrast-enhanced CT
LDCT	low dose CT
CTAP	CT arterial portography
CTHA	CT hepatic arteriography
LITS	Liver Tumor Segmentation Benchmark
HD	Hausdorff distance
VS	volume similarity
AD	absorbed dose
GELU	Gaussian Error Linear Unit

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